
**Experiment name: HDL(Verilog) Simulation Of Logic Gates Using
ModelSIM**



Department of Electrical and Electronic Engineering

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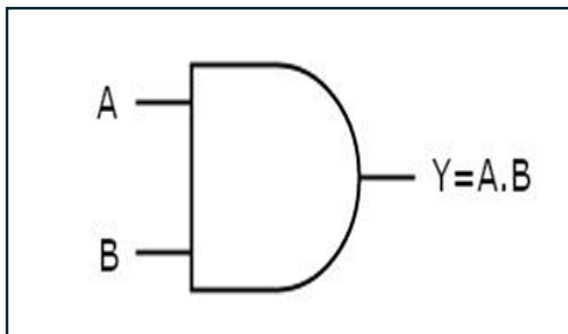
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Theory:

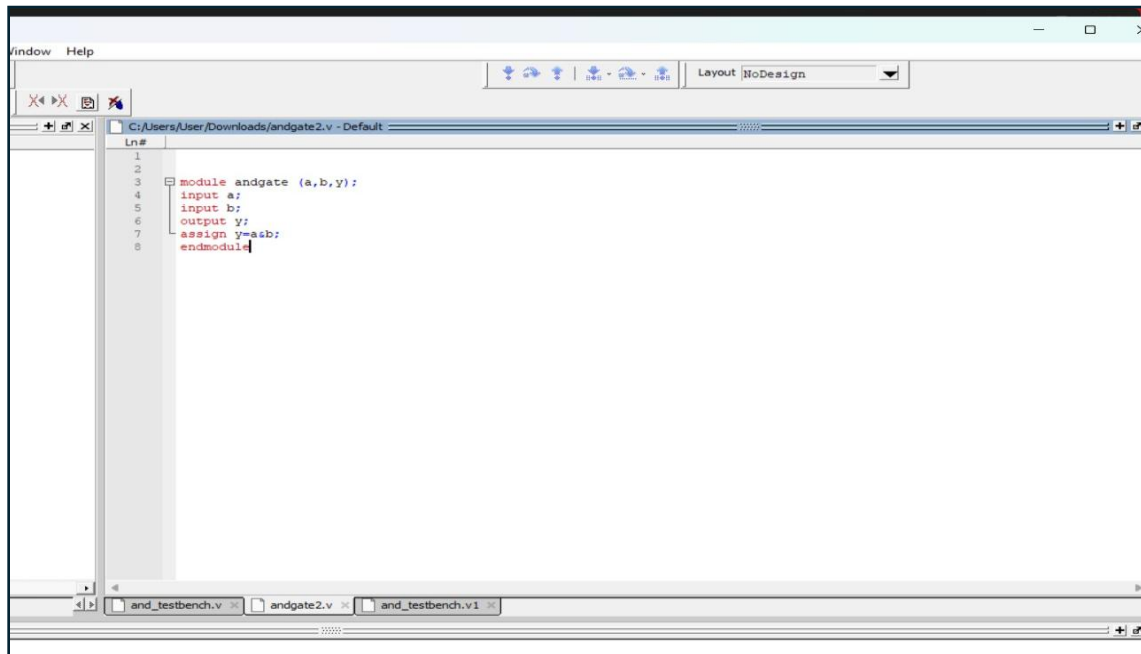
Hardware Description Language (HDL) is a special type of language used to describe and design digital electronic circuits. Among different HDLs, verilog is widely used because its syntax is relatively simple and it allows designers to test a circuit before implementing it in actual hardware. In this experiment, verilog HDL is used to design a basic AND gate, and its operation is verified through ModelSim simulation. An AND gate produces a high (1) output only when both of its inputs are high. Its Boolean expression is $Y = A \cdot B$, where A and B are the input signals and Y is the output. If any one of the inputs is low (0), the output becomes low. To check the designed circuit, a testbench is created to apply all possible combinations of the two inputs. modelsim is then used to compile the Verilog program, run the simulation, and observe the changes in the input and output signals through waveforms. By comparing the simulated results with the AND gate truth table, we can easily confirm whether the designed circuit is working correctly. This experiment also gives a basic understanding of HDL-based digital circuit design and simulation.



A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

Fig:input and gate logic diagram and input and gate truth table

Verilog code:

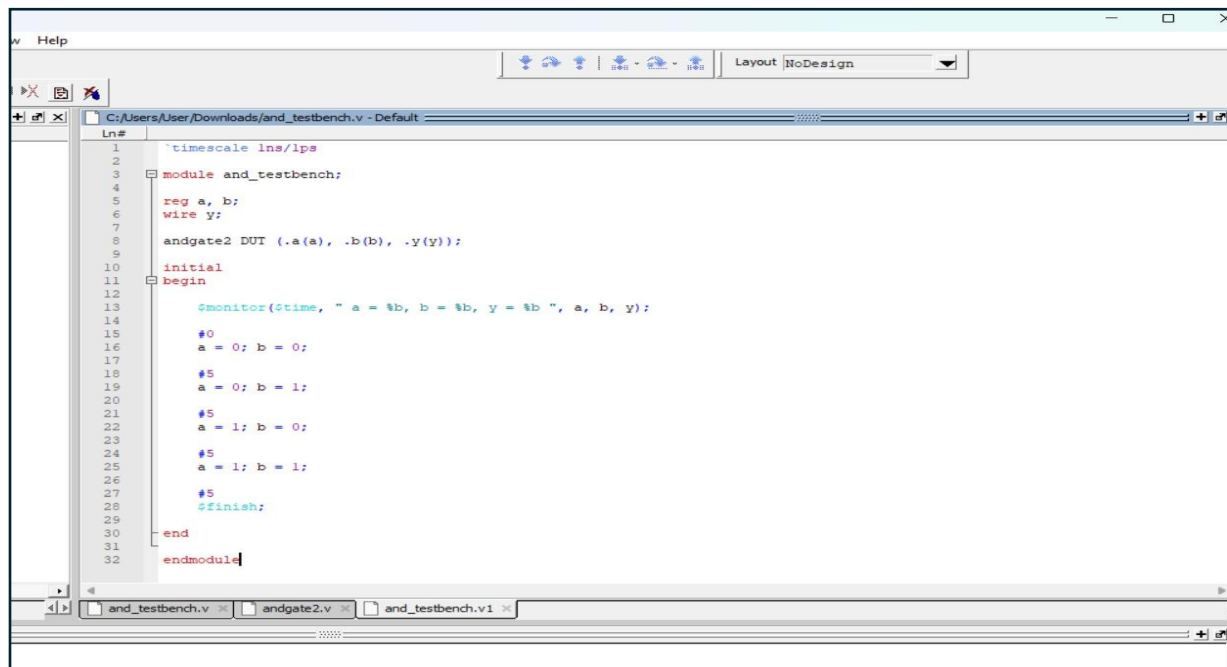


The screenshot shows a Verilog code editor window titled "C:/Users/User/Downloads/andgate2.v - Default". The code defines a module named "andgate" with two inputs, "a" and "b", and one output, "y". The output "y" is assigned the value of "a" AND "b" using the "assign" statement. The code is as follows:

```
1  
2  
3 module andgate (a,b,y);  
4   input a;  
5   input b;  
6   output y;  
7   assign y=a&b;  
8 endmodule
```

The editor interface includes a menu bar with "Window" and "Help", a toolbar with various icons, and a status bar at the bottom showing the current layout as "NoDesign".

Fig : Verilog code for 2 input and gate



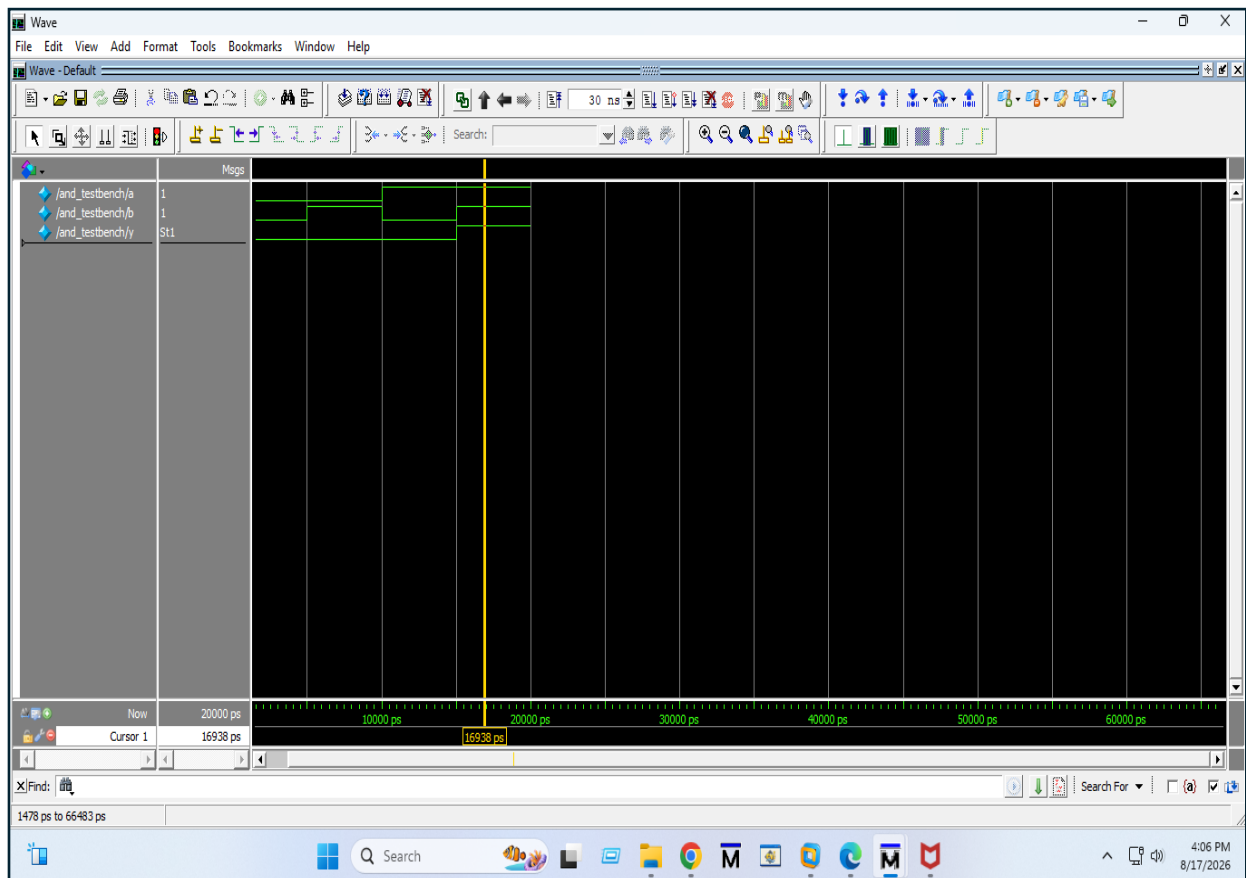
The screenshot shows a Verilog code editor window titled "C:/Users/User/Downloads/and_testbench.v - Default". The code defines a testbench module named "and_testbench" that instantiates the "andgate2" module. The testbench includes a time scale of 1ns/1ps, registers for inputs "a" and "b", and a wire for output "y". It uses a "monitor" function to display the values of "a", "b", and "y" at each time step. The code is as follows:

```
1 timescale 1ns/1ps  
2  
3 module and_testbench;  
4   reg a, b;  
5   wire y;  
6  
7   andgate2 DUT (.a(a), .b(b), .y(y));  
8  
9   initial  
10  begin  
11    $monitor($time, " a = %b, b = %b, y = %b ", a, b, y);  
12  
13    #0  
14    a = 0; b = 0;  
15  
16    #5  
17    a = 0; b = 1;  
18  
19    #5  
20    a = 1; b = 0;  
21  
22    #5  
23    a = 1; b = 1;  
24  
25    #5  
26    $finish;  
27  end  
28 endmodule
```

The editor interface is similar to the first screenshot, with a menu bar, toolbar, and status bar.

Fig : 2 input testbench Verilog code

Result:



Discussion :

In this experiment, a two-input AND gate was designed using Verilog HDL and simulated in ModelSim. The main purpose was to understand how a simple digital logic circuit can be described through HDL instead of constructing it physically with logic ICs. The Verilog statement `assign y = a & b;` represents the Boolean expression of the AND gate, where the output becomes HIGH only when both inputs are HIGH. A separate testbench was used to apply all four possible combinations of inputs: 00, 01, 10, and 11. The simulation waveform clearly showed that the output remained LOW for the first three combinations and became HIGH only for the 11 combination. This result matched the theoretical truth table of the AND gate. Through this experiment, I also understood the practical role of a testbench and how ModelSim can be used to observe signal changes with

In this experiment, I designed a simple two-input AND gate using Verilog HDL and tested it in ModelSim. The experiment

helped me understand how a basic digital logic circuit can be written as a program and then checked through simulation. I used the Verilog statement `assign y = a & b;` to describe the AND operation. To check whether the circuit was working correctly, I created a testbench and applied all four possible combinations of the two inputs. During the simulation, I observed that the output remained 0 when either of the inputs was 0. When both inputs became 1, the output also became 1. The waveform in ModelSim clearly showed this change, which matched the expected truth table of an AND gate. While doing the experiment, I also became familiar with the basic purpose of a testbench, the use of input and output signals, and how wave

Conclusion:

In conclusion, this experiment helped me understand the basic process of designing and simulating a digital logic gate using Verilog HDL and ModelSim. I successfully implemented a two-input AND gate and tested it with all possible combinations of input signals. The simulation waveform showed that the output became 1 only when both inputs were 1, which matched the expected truth table. Through this experiment, I also became familiar with writing a simple Verilog module, creating a testbench, and observing the output using the ModelSim waveform window. Overall, the experiment gave me a clear practical idea of how HDL can be used to design and verify digital circuits before implementing them in actual hardware.